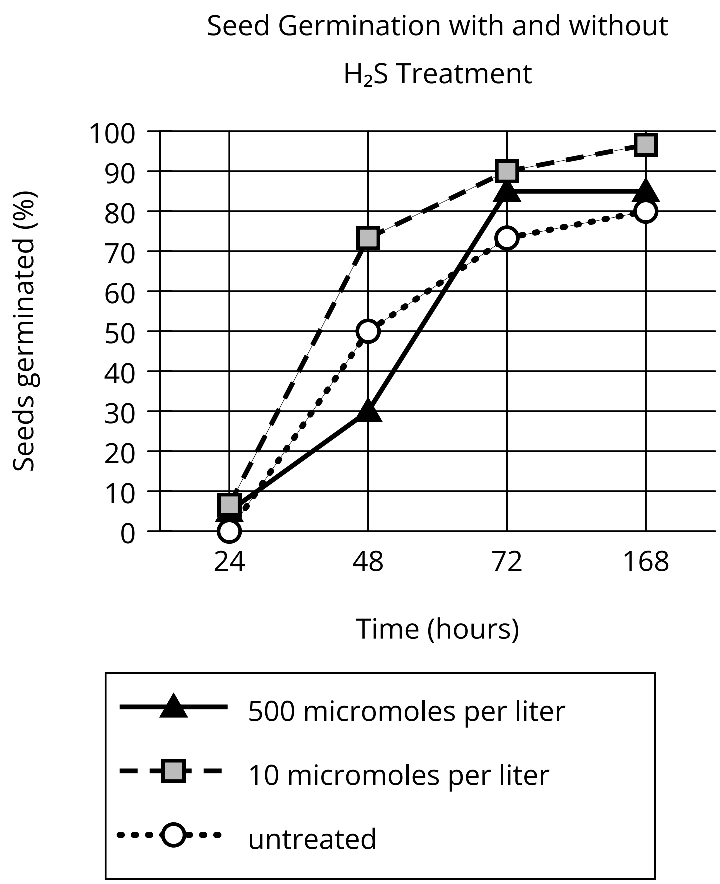
Question ID: 8a584241

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Medium

Question



In high concentrations, hydrogen sulfide (H_2S) is typically toxic to many plants. Frederick D. Dooley and colleagues wanted to understand what effects low doses of H_2S might have on plant growth. They treated bean, corn, wheat, and pea seeds with various concentrations (measured in micromoles per liter) of H_2S and tracked the germination of those seeds along with the germination of untreated seeds. Treatment with particular concentrations of H_2S was associated with accelerated germination: for example, ______

Which choice most effectively uses data from the graph to complete the statement?

Answer

A. at 24 hours, less than 10% of seeds treated with H_2S at a concentration of 10 micromoles per liter had germinated, whereas more than 90% of those seeds had germinated at 168 hours.

B. at 48 hours, more than 70% of seeds treated with H_2S at a concentration of 10 micromoles per liter had germinated, whereas only approximately 50% of untreated seeds had germinated.

C. at 168 hours, more than 90% of seeds treated with H_2S at concentrations of 10 or 500 micromoles per liter had germinated, whereas less than 70% of untreated seeds had germinated.

D. at 48 hours, approximately 50% of seeds treated with H_2S at a concentration of 10 micromoles per liter had germinated, whereas only approximately 30% of untreated seeds had germinated.

Correct Answer: B

Rationale

Choice B is the best answer. The claim is that some concentrations of H_2S led to increased germination rates, and this choice accurately shows that seeds treated with 10 micromoles per liter of H_2S tended to germinate faster than untreated seeds.

Choice A is incorrect. This choice doesn't justify the claim. The claim compares the germination rates of seeds exposed to certain concentrations of H_2S to untreated seeds, but this choice only discusses one concentration of H_2S, so it can't support any comparison between treated and untreated groups. Choice C is incorrect. This choice misreads the graph. At 168 hours, only about 85% of seeds treated with H_2S at 500 micromoles per liter and well over 70% of untreated seeds had germinated (about 80%). Choice D is incorrect. This choice misreads the graph. At 48 hours, about 70% of seeds treated with H_2S at 10 micromoles per liter and about 50% of untreated seeds had germinated.